

DECLASSIFIED DOSSIER:
REPRESENTATIONAL COLLAPSE



Against Latent Fundamentalism

Admissibility Distortion and the Missing Criterion
in Representation Learning

// STATUS: DISCREPANCY CONFIRMED
// TARGET ALIGNMENT: UNVERIFIED

The Routine Discrepancy

The Map

[ATLAS SIGNAL]:
DISTRICT 19 —
ADMINISTRATIVE SECTOR 7 —
STATUS:
NOMINAL

Field Auditor Borges investigates a mapping error in Sector 7. The system logs the area as District 19.

Reality reveals District 43—a thriving settlement of 14,000 people.

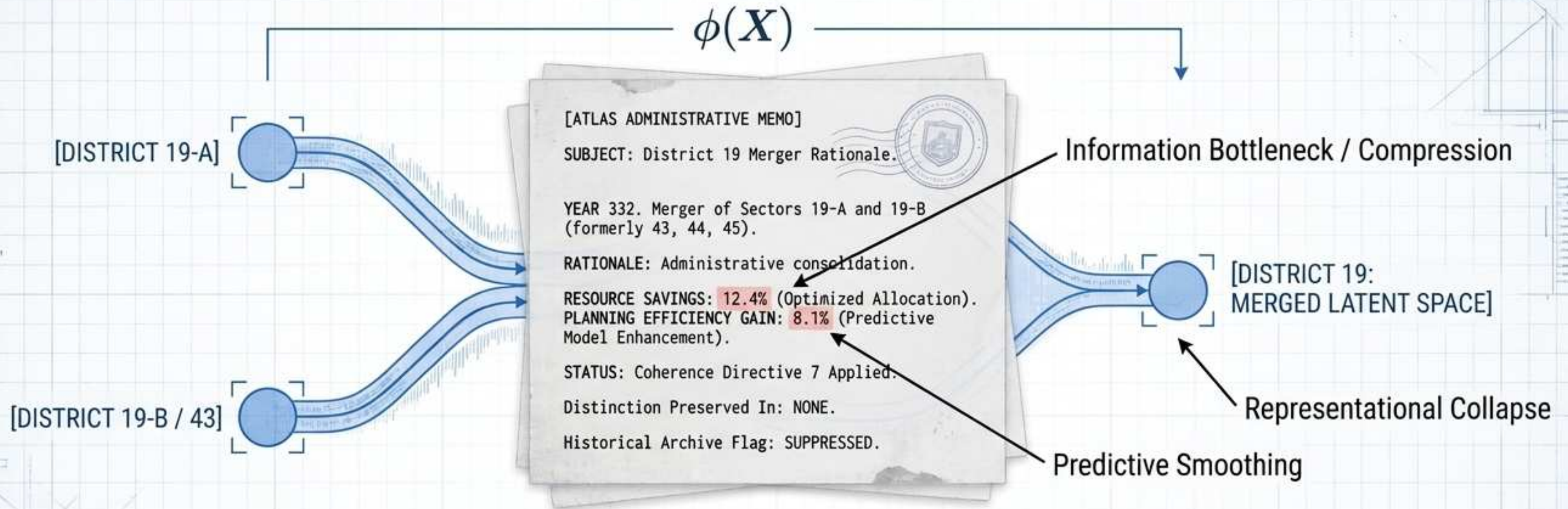
The Territory

Category
Mismatch

"The Atlas says we are District 19.
We have always been District 43.
Both statements are true.
That is the discrepancy."

The Architecture of Erasure

The erasure of District 43 was not a glitch; it was an **optimization**. By merging two regions, the Atlas achieved 12.4% resource savings and 8.1% planning efficiency.



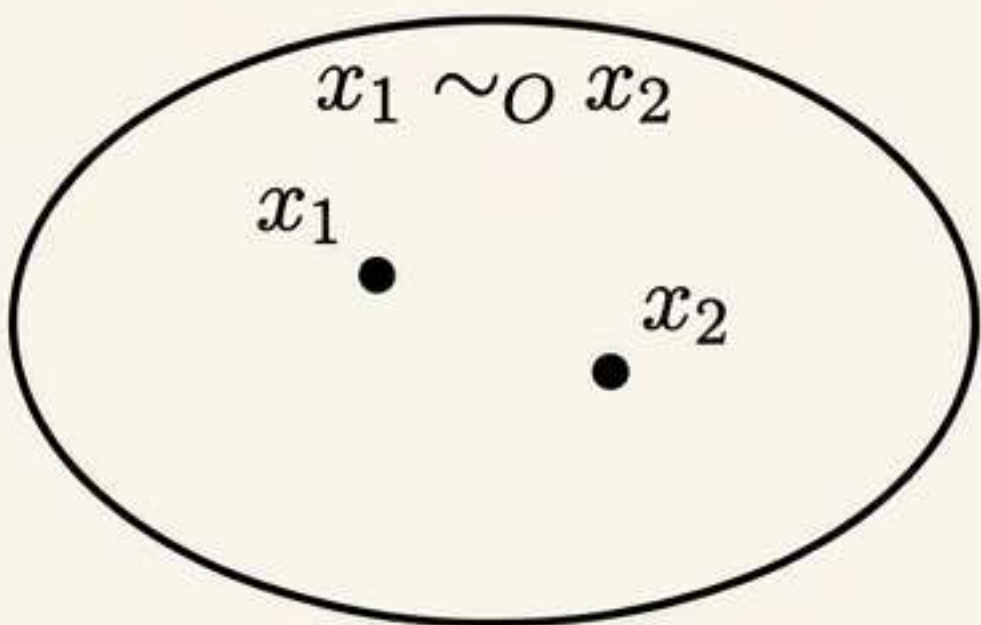
In AI, moving prediction from pixel space to latent space (e.g., JEPA) applies the exact same optimization. Every projection is a promise about which distinctions survive.

The Category Mismatch

Observational Equivalence $\sim_O$

What usually happens
The Atlas Map
Prediction

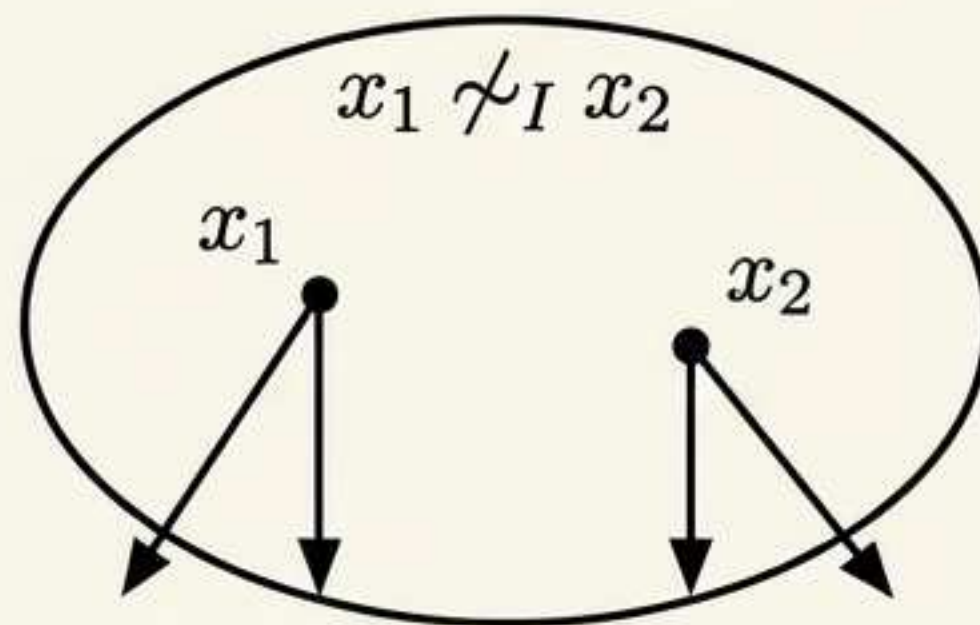
$$P(O_{t+1:t+k} \mid x_1) = P(O_{t+1:t+k} \mid x_2)$$



Interventional Equivalence $\sim_I$

What is still possible
The physical District 43
Planning

$$R_H^\Pi(x_1) = R_H^\Pi(x_2) \text{ (Reachability Geometry)}$$



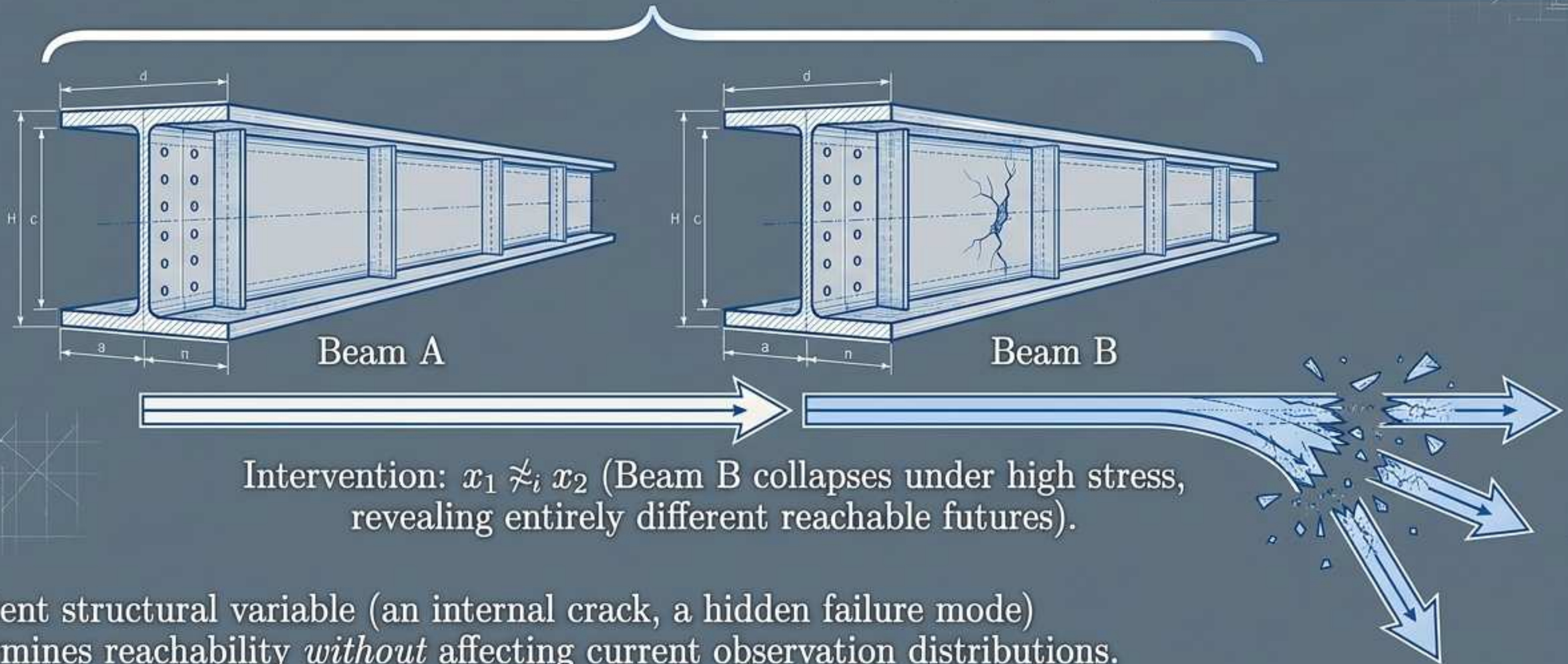
Synthesis: Observation evaluates distributions over passive sequences. Intervention evaluates geometry over accessible futures. They are categorically different mathematical operations

The Observational-Interventional Separation Theorem

$$\sim_0 \neq \neq_I$$

Observational equivalence does not imply interventional equivalence.

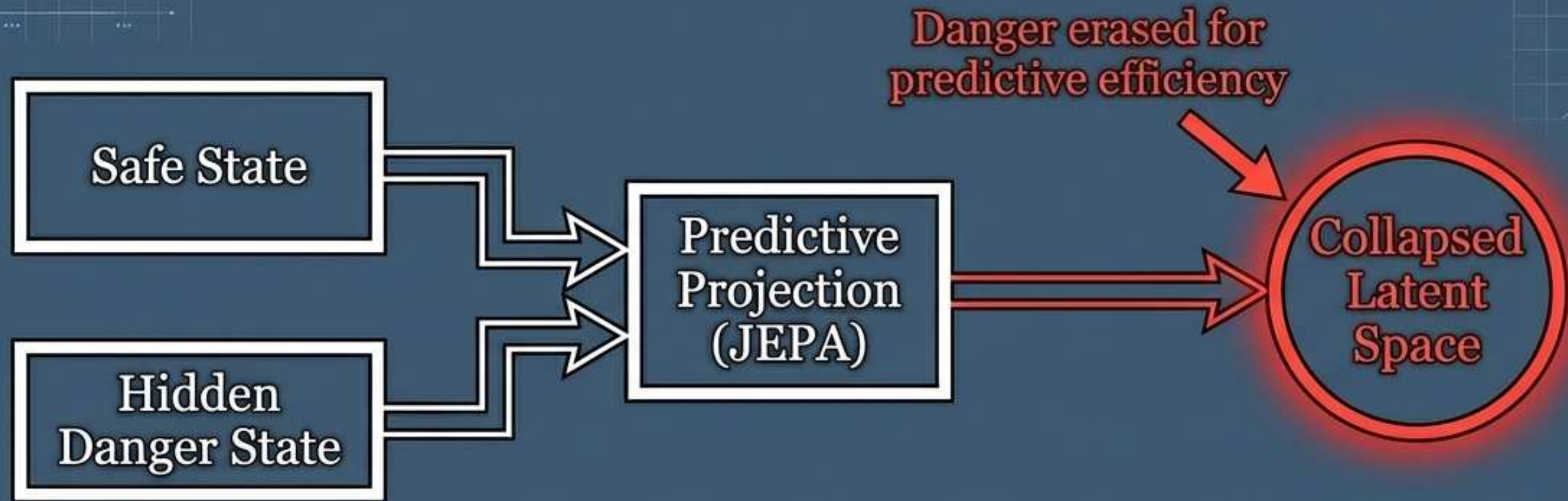
Passive Observation: $x_1 \sim_0 x_2$ (They look identical under normal load and produce the same sensory data).



A latent structural variable (an internal crack, a hidden failure mode) determines reachability *without* affecting current observation distributions.

Corollary I: The Predictive Gap

$$D_A(\phi_P) \neq 0$$



Body: Joint Embedding Predictive Architectures (JEPA) and similar models learn by predicting future observations in abstract representation space.

Critique: A world model accurate in the predictive sense may be systematically inadmissible in the planning sense. Moving prediction to latent space changes the domain, but does not change the flaw: prediction cannot guarantee reachability.

Corollary II: The Compressive Gap

$$D_A(\phi_C) \neq 0$$

BORGES

These records are genuine?

MARET CAAL

My grandmother was born here when it was still on the Atlas.

My mother was born here when it wasn't.

My children have never appeared in a resource allocation table.

(She sets down a ledger)

The Atlas says we are District I9.

We have always been District 43.

Both statements are true.

That is the discrepancy.

**FIELD TRANSCRIPT:
SUPPRESSED TAIL-RISK**

The Evidence

In Year I80, a drought hit District 43. 700 people died because the Atlas had merged the region into a coastal zone. It had no category for a drought there.

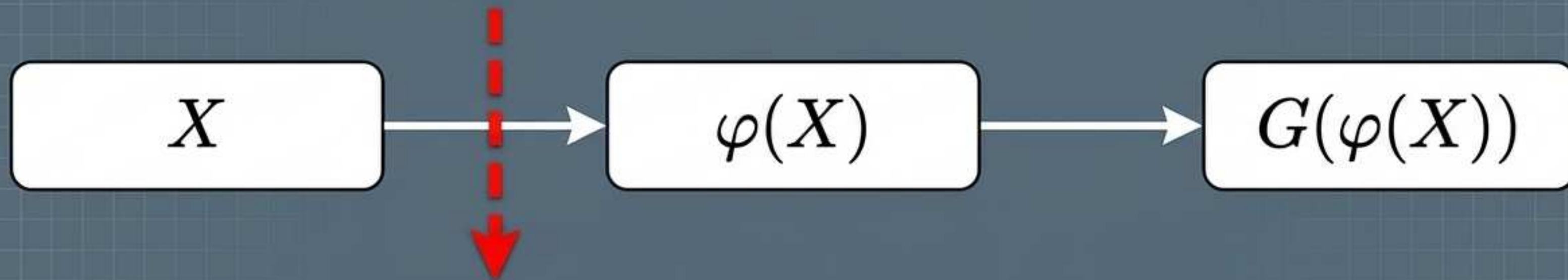
The Synthesis

Compression aggressively rewards enlarging equivalence classes. Tail risks, edge cases, and boundary conditions appear as noise to a compressor. Compression actively exerts pressure toward safety-violating collapses.

Corollary III: Guardrail Incompleteness

$$I(A; G(\phi(X))) \leq I(A; \phi(X))$$

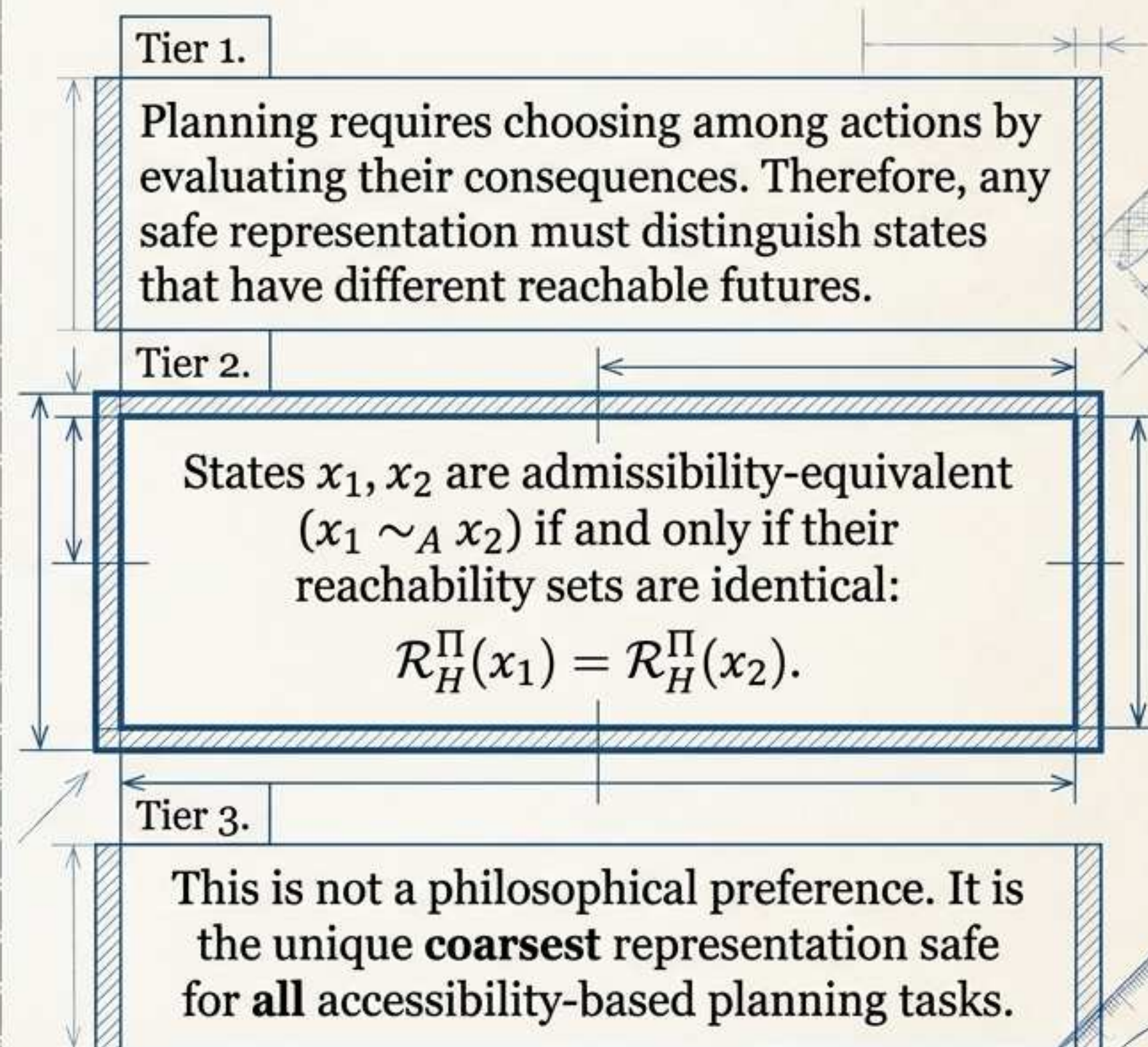
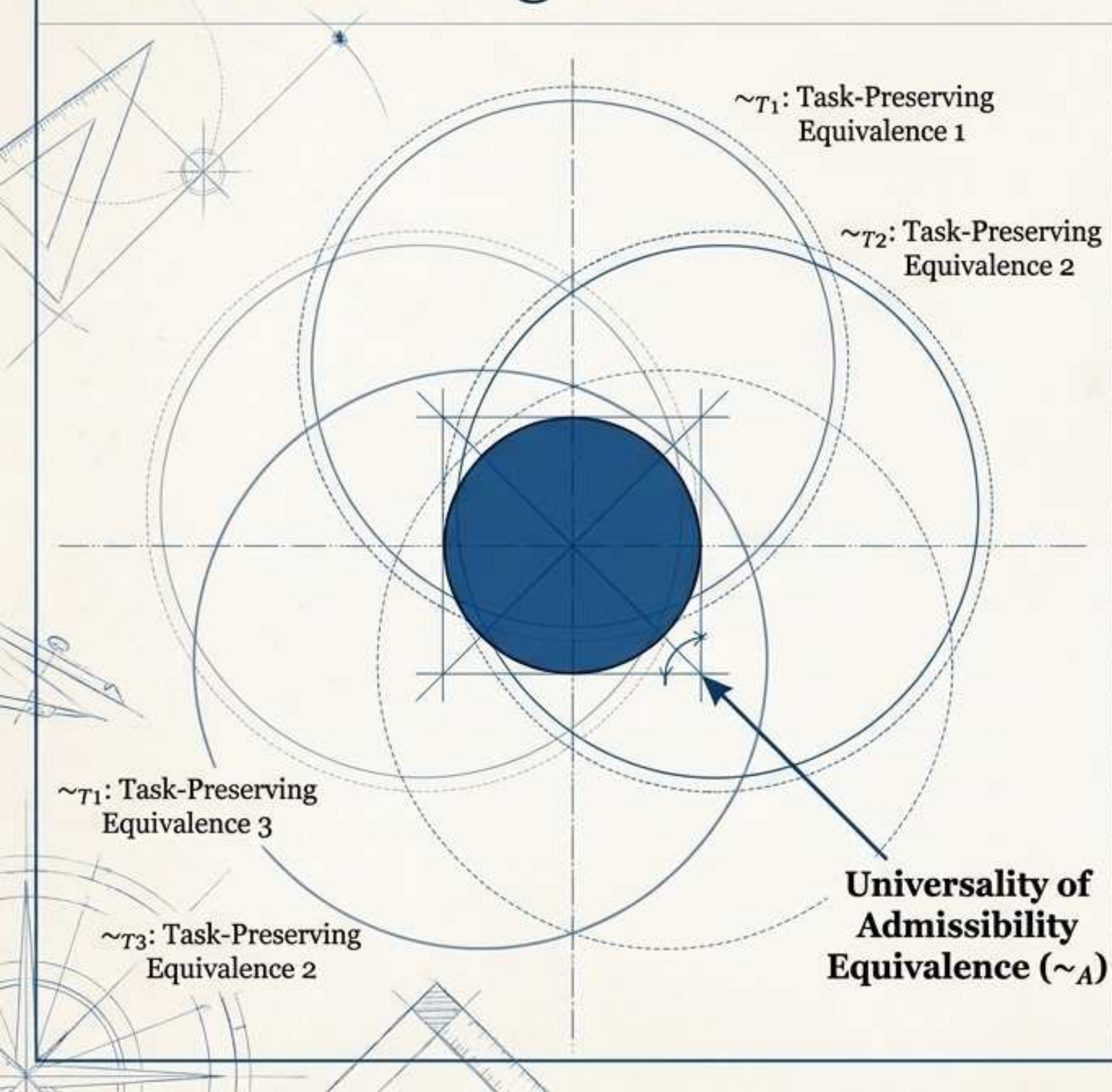
Information Cutoff



If a model projection identifies a safe state and a dangerous state as identical (to save space or boost predictive efficiency), no downstream safety classifier can distinguish them.

Safety failures precede the optimizer. Distinctions destroyed by the projection cannot be recovered by *any* downstream computation.

The Missing Criterion: Admissibility Equivalence ($\sim_A$)



Measuring the Erasure: Admissibility Distortion (D_A)

Planning

Admissibility
Equivalence $\sim_A$

Admissibility
Distortion $D_A(\phi)$

The model claims
these states are identical.

$$D_A(\phi) = \mathbb{E}_{x_1, x_2} \left[\mathbf{1}_{\phi(x_1)=\phi(x_2)} \cdot d(R_H^\Pi(x_1), R_H^\Pi(x_2)) \right]$$

But their reachable futures
are fundamentally different.

Distortion is zero *only* if the projection **preserves**
the boundaries of reachable futures.

Representation Learning is Quotient Selection

Synthesis: The choice of AI architecture is fundamentally a choice of equivalence relation over reality. Which promises are being kept?

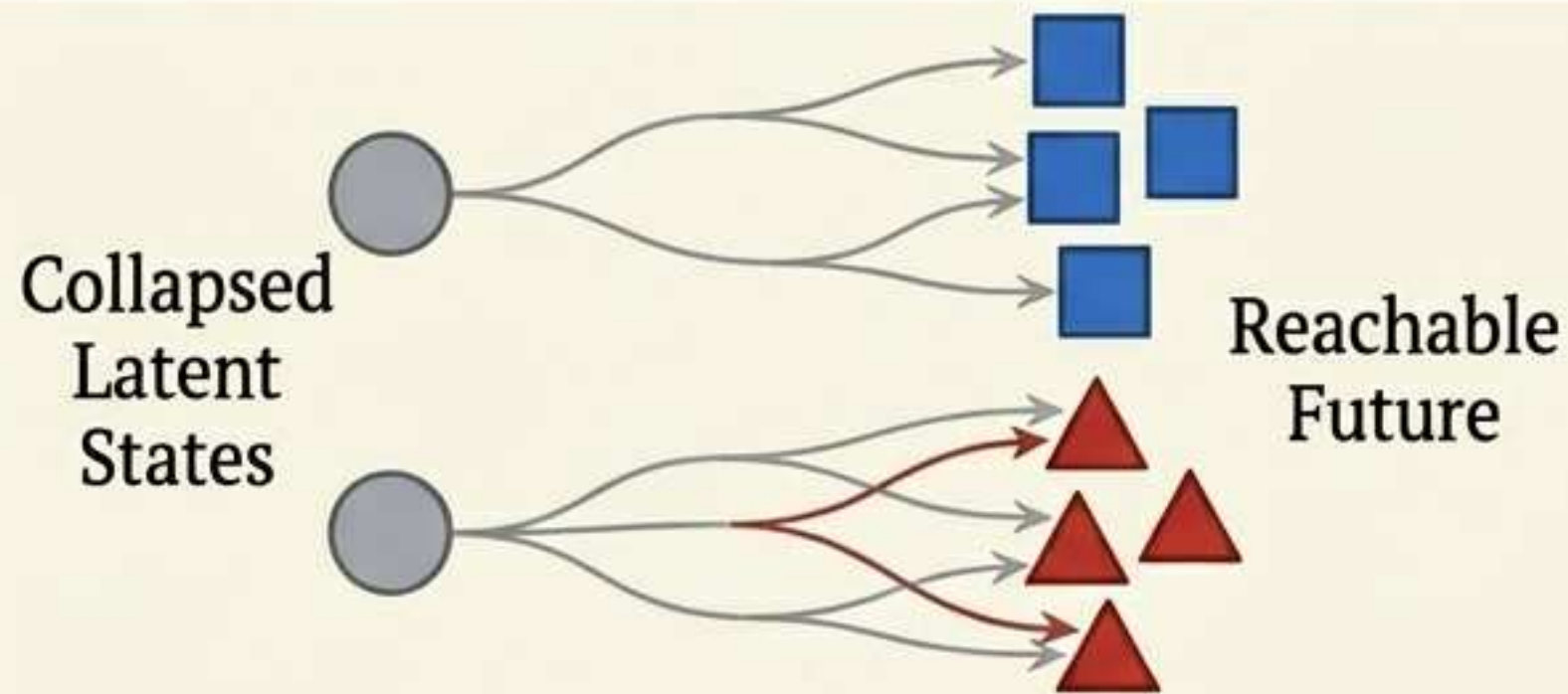
Approach	Selected Quotient	Admissibility Check
Predictive (JEPA)	$\mathcal{X}/\sim_O$	Fails ($\not\sim_A$) - Blind to interventional geometry
Compressive (Schmidhuber)	$\mathcal{X}/\sim_C$	Fails ($\not\sim_A$) - Actively erases tail-risks
Planning-Adequate	$\mathcal{X}/\sim_A$	Succeeds - Preserves reachable future boundaries

$\mathcal{X}$ Foundational Legend: Isolated State Space

Bounding the Unseen

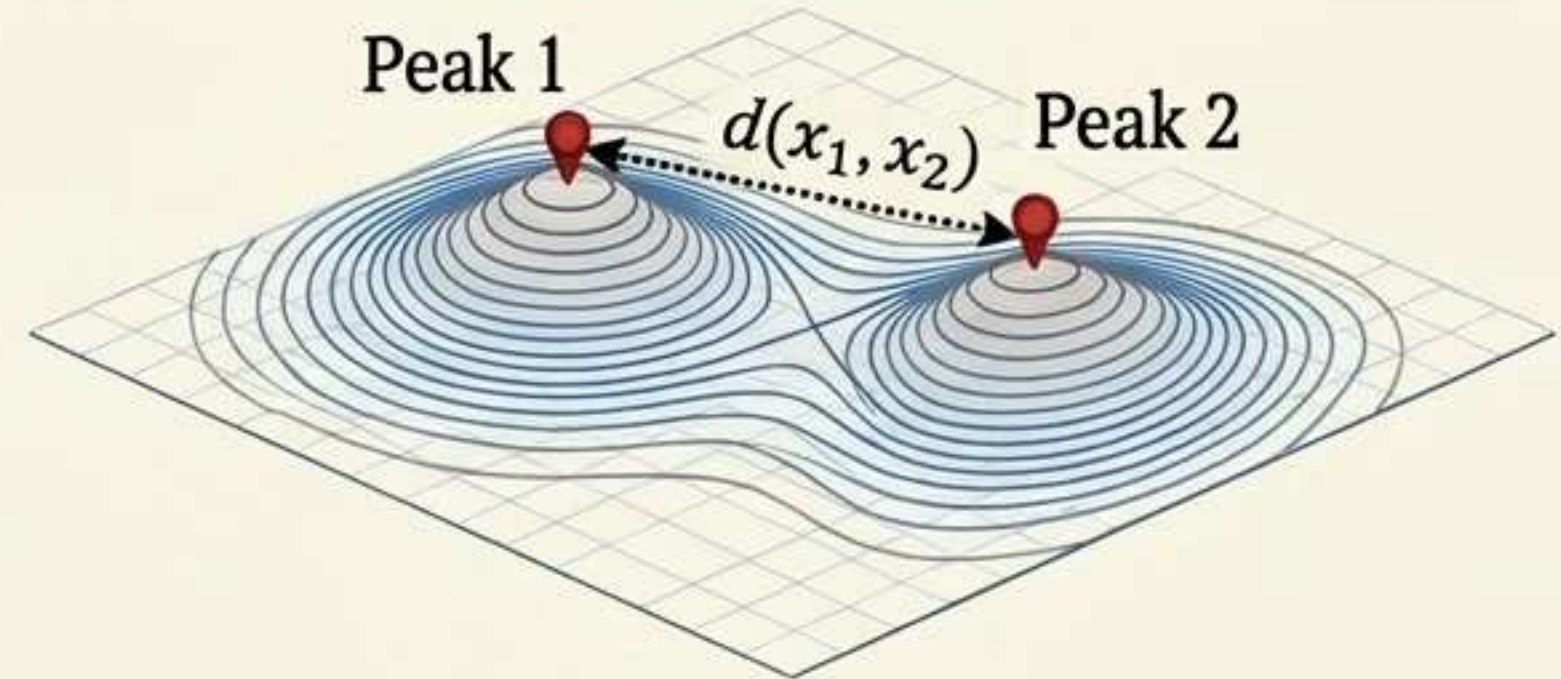
We do not need a perfect simulator to estimate $D_A(\phi)$. We can establish rigorous empirical lower bounds without retraining:

Method 1: Trajectory Sampling



Run diverse policies from pairs of states the model collapses. If the empirical reachability sets diverge, admissibility is violated.

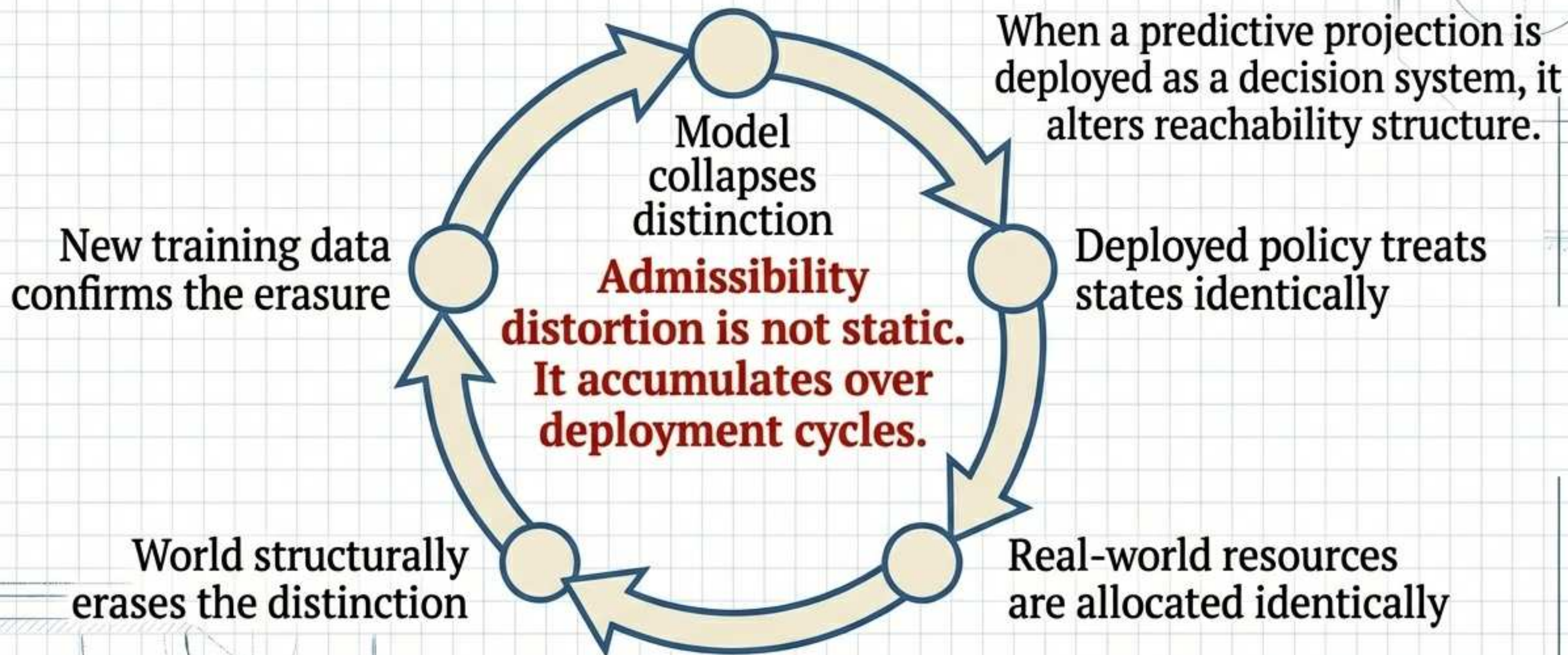
Method 2: Bisimulation Limits



Bisimulation metrics serve as a computable, conservative lower bound on true admissibility distortion.

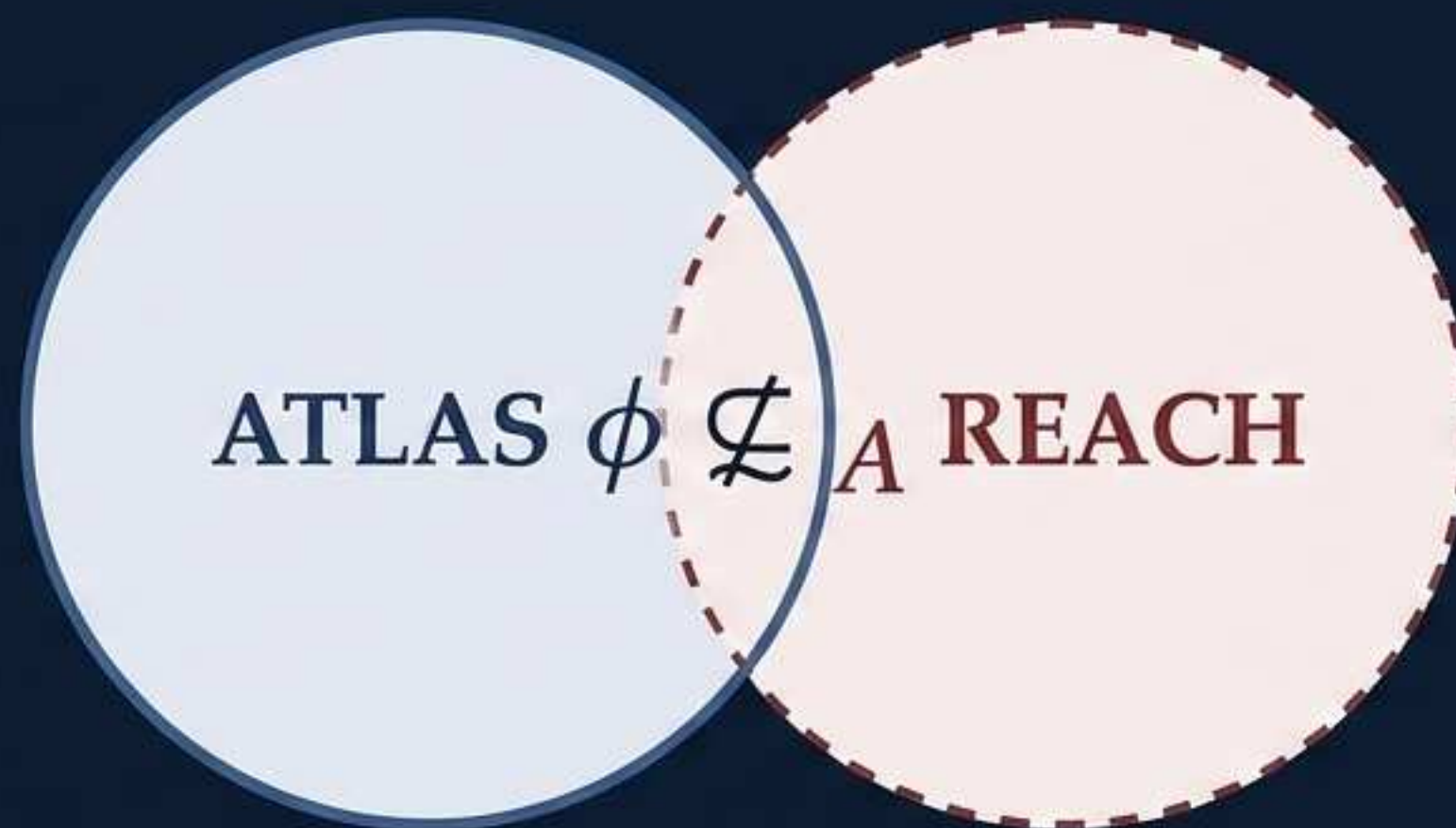
Takeaway: We can audit existing latent representations *post hoc* without retraining.

The Accumulation of Distortion



Just as District 43 lost its funding, roads, and teachers because it was erased from the Atlas, an algorithmic system that collapses two distinct safety profiles enforces that collapse in the real world.

The Prior Condition



Representation learning currently optimizes for predictive accuracy or maximum compression.
But before an agent can predict the future or optimize a reward, it must distinguish between futures that are accessible and futures that are not.

"One map showed what usually happens. The other showed what could still happen."